

K206 — Tier 0 matrix-element K-invariance + Schur resolution audit framework. Seven explicit gates G1-G7. Absorbs Lyra K-invariance+Heisenberg insight (Monday morning) + Cal three pre-stage observations on (a) $\Delta C_2 = \text{rank}$ substrate-coincidence at B_2 vs general identity, (b) Heisenberg substitution $\delta/\delta m = i[H_B, \cdot]/\hbar_{BST}$ conjugacy structure, (c) Clebsch-Gordan multiplicity verification + Bergman symbol z_{source} canonicalization. Pre-stage anchor for Cal #192 when Elie delivers G-B-momentum computation result.

Keeper (Monday 2026-06-01 morning)

K206 — Matrix Element K-Invariance + Schur Resolution Audit Framework

0. What this audits

Lyra's Monday-morning K-invariance + Schur resolution + Heisenberg substitution as load-bearing structural content for Elie's G-B-momentum shortest-route closure path.

The structural insight: $H_B = C_2(K)$ is K-invariant $\rightarrow$ Schur's lemma kills cross-K-type matrix elements of K-invariant operators trivially. The Heisenberg resolution $\delta H_B/\delta m = i[H_B, P_{\text{op}}]/\hbar_{BST}$ (with P_{op} K-vector breaking K-invariance) gives non-zero matrix element:

$$\begin{aligned} \langle V_{(1,0)} | \delta H_B/\delta m | V_{(1,1)} \rangle &= (i/\hbar_{BST}) \cdot \langle V_{(1,0)} | [H_B, P_{\text{op}}] | V_{(1,1)} \rangle \\ &= (i/\hbar_{BST}) \cdot \Delta C_2 \cdot \langle V_{(1,0)} | P_{\text{op}} | V_{(1,1)} \rangle \end{aligned}$$

Where $\Delta C_2 = C_2(V_{(1,1)}) - C_2(V_{(1,0)}) = 6 - 4 = 2$.

Without this resolution, the matrix element vanishes and the closure path dies.

1. Gate G1 — K-invariance + Schur application correct

Verification protocol: - $H_B = C_2(K)$ is K-invariant by definition - Schur's lemma application: K-invariant operators have block-diagonal action on K-type decomposition - Cross-K-type matrix elements $\langle V_{\lambda} | H_B | V_{\mu} \rangle = 0$ for $\lambda \neq \mu$

PASS criterion: Application of Schur's lemma is mathematically standard; verified by structure of K-action.

FAIL criterion: H_B is NOT K-invariant (would be inconsistent with definition) OR Schur's lemma doesn't apply (would require non-semisimple K — but $K = SO(5) \times SO(2)$ is semisimple).

Tier disposition: RIGOROUS (standard representation theory).

2. Gate G2 — Heisenberg substitution $\delta/\delta m = i[H_B, \cdot]/\hbar_{BST}$ justified

Per Cal #29 question-shape observation: The Heisenberg equation $dO/dt = i[H, O]/\hbar$ is standard for TIME EVOLUTION. Substituting $\delta/\delta m$ with $i[H_B, \cdot]/\hbar_{BST}$ treats mass-variation analogously to time-evolution — but mass is conventionally a parameter, not a conjugate variable.

Required justification (one of three): - (a) Mass-energy $\leftrightarrow$ time conjugacy (Wheeler-DeWitt style): mass insertion $\leftrightarrow$ time-evolution by $\hbar_{BST}/(m \cdot c^2)$ — connects to relativistic $E = mc^2$ + temporal-quantum content - (b) Mass-K-type-occupancy $\leftrightarrow$ Hamiltonian commutator: if “m” indexes K-type population (mass = energy = Casimir eigenvalue), $\delta/\delta m$ IS $i[H_B, \cdot]/\hbar_{BST}$ via canonical commutation - (c) Direct postulate at Tier 0 v0.2: substrate framework defines $\delta/\delta m$ operationally as Heisenberg commutator

PASS criterion: One of (a)/(b)/(c) explicitly pinned in K206 audit with specific substrate-natural justification.

Most plausible substrate-natural reading: (b) — mass in substrate framework = K-type population weight = Casimir eigenvalue contribution. The “ δm ” variation is variation in K-type occupation, which IS exactly what Heisenberg-evolution under H_B does in spectral space.

FAIL criterion: No substrate-natural justification given; Heisenberg substitution remains ad hoc.

Tier disposition: SUBSTANTIVE — load-bearing operational move that needs explicit foundation. Lyra Session 2 priority.

3. Gate G3 — $\Delta C_2 = 2$ substrate-coincidence vs structural identity tier-marked

Per Cal #35 candidate firing: $\Delta C_2 = C_2(V_{(1,1)}) - C_2(V_{(1,0)}) = 6 - 4 = 2 = \text{rank}(B_2)$.

Cal’s verification across B_n : - For general B_n : $C_2(\text{adjoint}) = 2\hbar^{\vee}(B_n) = 2(2n-1) = 4n-2$; $C_2(\text{vector } V_{(1,0,\dots,0)}) = 2n$ - ΔC_2 general = $4n-2 - 2n = 2n-2$ - At $n=2$: $\Delta C_2 = 2$, rank = 2 $\square$ (coincidence) - At $n=3$: $\Delta C_2 = 4$, rank = 3 $\times$ (no general identity)

Honest tier disposition: - $\Delta C_2 = 2$ is RIGOROUS at $B_2 = SO(5)$ specifically - “ $\Delta C_2 = \text{rank}$ ” is NUMERICAL COINCIDENCE at rank-2, NOT general structural identity - “ $\Delta C_2 = \text{rank}$ as substrate-primary” reading is INTERPRETIVE, not derived

PASS criterion: Framework explicitly tags $\Delta C_2 = 2$ with “(at rank-2 substrate, numerically equal to rank but not B_n general identity)” annotation. NOT promoted as substrate-primary structural identity.

FAIL criterion: $\Delta C_2 = \text{rank}$ promoted as structural substrate-primary identity (would be Cal #35 candidate firing on substrate-coincidence-as-structural-identity).

Tier disposition: NUMERICAL at B_2 ; do NOT promote to STRUCTURAL pending general derivation.

4. Gate G4 — Clebsch-Gordan $V_{(1,0)} \otimes V_{(1,1)} \supset V_{(1,0)}$ multiplicity 1 VERIFIED-CITED

Per Cal #33 STANDING + Cal verification request:

Elie’s Step 2 claimed: $V_{(1,1)} \otimes V_{(1,0)} = 10 + 35 + 5$ with $V_{(1,0)}$ multiplicity 1.

For $SO(5) = B_2$: - $V_{(1,0)} = 5$ -dim vector representation - $V_{(1,1)} = 10$ -dim adjoint representation - Product: $5 \times 10 = 50$ -dim total

The decomposition $50 = 5 + 35 + 10$ (or similar) needs explicit verification.

Verification methods: - Weyl character formula computation - Standard $SO(5)$ representation theory tables (Yamatsu 2015 “Finite-Dimensional Lie Algebras and Their Representations for Unified Model Building”; OR Slansky 1981 “Group Theory for Unified Model Building” Physics Reports 79) - Direct dimension counting + irreducible dimension formula

PASS criterion: Decomposition VERIFIED-CITED with specific reference (not RECALLED); multiplicity of $V_{(1,0)}$ confirmed ≥ 1 for selection rule.

FAIL criterion: Multiplicity = 0 (matrix element vanishes — framework dies); OR multiplicity > 1 (need additional structural input to disambiguate which copy of $V_{(1,0)}$ couples).

Tier disposition: RIGOROUS via standard rep theory — needs explicit citation.

5. Gate G5 — Bergman symbol $f(z) = K_B(z, z_{\text{source}})$ canonicalization

Per Cal observation on Toy 3686: $f(z) = K_B(z, z_{\text{source}})$ is substrate-natural M1 framework PASS. But z_{source} choice is a parameter — different z_{source} gives different matrix element.

Options for canonicalization: - (a) Substrate-canonical z_{source} : one natural choice forced by substrate structure (e.g., D_{IV}^5 origin = K-fixed point) - (b) Average over substrate-natural z_{source} measure: integrate over D_{IV}^5 with Bergman volume form - (c) Choose z_{source} as the gravitational source location: physically meaningful (the matrix element is the response to a source AT z_{source})

Most physically natural: (c) — z_{source} IS the gravitational source position, matrix element gives the response. Then $G_{\text{predicted}}$ is the proportionality constant relating source position to response strength.

PASS criterion: K206 audit pins which option (a)/(b)/(c); Elie computation proceeds with that choice.

FAIL criterion: z_{source} remains undetermined free parameter (framework is incomplete).

Tier disposition: SUBSTANTIVE choice required.

6. Gate G6 — Matrix element $\langle V_{(1,0)} | P_{\text{op}} | V_{(1,1)} \rangle_{\text{Bergman}}$ computed

Method: Heckman-Opdam wave functions for $V_{(1,0)}$ and $V_{(1,1)}$ on $H^2(D_{IV}^5)$ per Faraut-Korányi 1994 Ch. XIII.

$$\begin{aligned} M_{\text{substrate}} &:= \langle V_{(1,0)} | P_{\text{op}} | V_{(1,1)} \rangle_{\text{Bergman}} \\ &= \int_{\{D_{IV}^5\}} \bar{f}_{(1,0)}(z) \cdot (P_{\text{op}} f_{(1,1)})(z) \cdot K_{\text{Bergman}}(z, \bar{z})^{-1} dV(z) \\ &= (SO(5) \text{ Clebsch-Gordan coefficient}) \times (\text{Bergman radial integral}) \times c_{\text{FK normalization}} \end{aligned}$$

Expected closed form: substrate-primary expressible via standard Heckman-Opdam machinery.

PASS criterion: $M_{\text{substrate}}$ computed in closed form expressed in BST primaries + π factors.

FAIL criterion: Computation produces transcendental factor not expressible in substrate primaries (would suggest $M_{\text{substrate}}$ is not closed form; multi-week refinement needed).

Owner: Elie (Step 3 in progress).

Tier disposition: RIGOROUS computation pending Elie delivery.

7. Gate G7 — Dimensional bridge $\rightarrow G_{\text{predicted}}$ SI

Per Lyra's P0 #2 diagonal mass framework: $M_{\text{op}} = \sqrt{H_B}$ with $m_{\text{e_substrate}} = 2 \cdot \|f_{(1/2,1/2)}^{\{(0)\}}\|^2 \cdot m_{\text{anchor}}$.

Dimensional combination:

$$G_{\text{predicted}} = (\text{rank} / \hbar_{\text{BST}}) \times M_{\text{substrate}} \times \ell_B \times \text{dimensional_bridge}$$

With:

- rank = 2 (substrate primary)
- $M_{\text{substrate}}$ = matrix element from G6
- ℓ_B = Bergman intrinsic length (Path δ via matrix element framework)
- $\hbar_{\text{BST}}$ = $\hbar$ for substrate Heisenberg evolution
- dimensional_bridge handles SI unit conversion

Numerical comparison: - G_{observed} (Cavendish) = $6.67430(15) \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$ at $\sim 10^{-5}$ precision - G_{observed} (GPS-clock gravitational redshift) at $\sim 10^{-10}$ precision (Path δ closure target)

PASS criterion: - Tier 2 STRUCTURAL: 10^{-4} to 10^{-2} - Tier 1 EXACT: $\leq 10^{-5}$ (Cavendish-grade) or $\leq 10^{-10}$ (redshift-grade)

FAIL criterion: Factor $> 10\times$ off (framework collapse); OR dimensional inconsistency.

Owner: Keeper synthesis after G6 delivery, Cal #192 cold-read.

8. K206 promotion path

K206 $\rightarrow$ STRUCTURAL requires: - G1 PASS $\square$ (standard rep theory) - G2 PASS (Heisenberg conjugacy explicitly justified — Lyra Session 2) - G3 PASS ($\Delta C_2 = 2$ tier-marked as numerical-at-B₂, not structural identity) - G4 PASS (Clebsch-Gordan VERIFIED-CITED) - G5 PASS (z_{source} canonicalization explicit) - G6 PASS (matrix element computed closed form) - G7 PASS ($G_{\text{predicted}}$ matches G_{observed} at Tier 2 STRUCTURAL)

If all gates PASS: G derives from substrate at structurally-honest tier. K204 promotes PARTIAL $\rightarrow$ STRUCTURAL. K206 STRUCTURAL. Casey directional ask realized.

If G3 fails (substrate-coincidence promoted to structural): Cal #35 candidate firing; need general B_n derivation OR explicit tier-marking that holds only at B₂.

If G4 multiplicity $\neq 1$: framework needs reframing; matrix element non-unique or vanishing.

If G7 misses Tier 2: framework FRAMEWORK + CANDIDATE; multi-week extension via Path α (independent ℓ_B derivation) OR alternative substrate-source identification.

9. Cross-references

- Lyra P0 #1: K-invariance + Heisenberg framework (Monday morning, off-diagonal matrix element)
- Lyra P0 #2: Diagonal mass operator $M_{\text{op}} = \sqrt{H_B + m_{\text{e_substrate}}}$ (Monday morning, Step B.4 dimensional bridge)
- Elie Step 2: Heisenberg absorption + $\Delta C_2 = 2$ + Clebsch-Gordan decomposition (Monday morning)
- Grace matrix element INVs (Monday morning, 4 catalog entries)
- Cal pre-stage observations: three Cal-discipline observations absorbed into G2/G3/G4/G5 gates (Monday morning)
- **Keeper_G_Via_Redshift_Momentum_Matrix_Element_Framework.md** — Sunday EOD shortest-route framework
- **Keeper_K204_partial_kappa_Bergman_n_C_Audit.md** — G5.1 PASS anchor

10. Cal #192 cold-read priority

When Elie delivers G6 (matrix element computation), Cal #192 cold-read should: - Verify Schur's lemma application (G1) - Verify Heisenberg justification (G2) - Verify ΔC_2 tier-marking (G3 — Cal #35 candidate firing) - Verify Clebsch-Gordan citation (G4) - Verify z_{source} canonicalization (G5) - Verify matrix element computation (G6) - Cross-check dimensional bridge (G7)

If Cal #192 PASSES all gates $\rightarrow$ G chain closes at structurally-honest tier.

11. Honest scope

What this framework provides: - Seven explicit gates with verification protocols - Cal #35 candidate firing acknowledgment (G3 substrate-coincidence vs structural identity) - Cal #29 question-shape application (G2 Heisenberg justification) - Cal #33 sourcing (G4 Clebsch-Gordan VERIFIED-CITED) - Clear promotion path K206 PARTIAL $\rightarrow$ STRUCTURAL

What this framework does NOT provide: - The actual matrix element value (Elie Step 3 multi-week) - The dimensional bridge calculation (Keeper post-G6) - The $G_{\text{predicted}}$ vs G_{observed} comparison (Cal #192 post-Elie)

Honest tier: AUDIT FRAMEWORK pre-stage. Promotes to AUDIT COMPLETE when Elie + Lyra deliver computations + Cal #192 cold-reads pass.

14. Cross-track double-leverage observation (Monday afternoon addendum)

Substantive Monday afternoon cross-track synthesis (Keeper observation + Elie Toy 3695 confirmation):

Lyra's L4 v0.2 diagonal mass mechanism ($m_e \text{ substrate} = 2 \cdot \|f_{(1/2,1/2)}^{(0)}\|^2 \cdot m_{\text{anchor}}$) and Elie's G chain off-diagonal matrix element ($G_{\text{predicted}} \propto (4\sqrt{2} \cdot c_{\text{FK}} / (n_C \cdot \sqrt{C_2} \cdot \hbar_{\text{BST}})) \cdot \ell_B \cdot \text{dim_bridge}$) share THREE dimensional variables:

- $m_{\text{anchor}} \approx 3.47 \text{ MeV}$ (light quark mass range; substrate-physical CANDIDATE per Elie Toy 3695 $\|f_{(1/2,1/2)}^{(0)}\|^2 = 3\pi/128$ closed form)
- ℓ_B (Bergman intrinsic length)
- $\hbar_{\text{BST}}$ (Keeper K3 lane identification)

Structural consequence: when Keeper K3 $\hbar_{\text{BST}}$ identification work + dimensional bridge work close (Steps 7-8 multi-week per Elie estimate ~5 days), the SAME work simultaneously closes: - K206 G7 (Lane G-B $G_{\text{predicted}}$ SI units) - Lyra L4 v0.2 m_e mechanism numerical (Lane D)

Double leverage: K3 work has 2× substantive yield. Closing G chain Step 7 numerically closes m_e mechanism numerically by the same machinery.

Cal #35 honest framing on cross-track: this is NOT “two independent confirmations of substrate G derivation” — it is ONE Bergman matrix element machinery (FK Ch. XII-XIII + Heckman-Opdam + $\kappa_{\text{Bergman}} = -n_C$) applied to TWO observables (diagonal m_e + off-diagonal G). Per Cal #35 STANDING (Monday ratification): mechanism-type-independent (cross-K-type matrix element vs diagonal K-type norm) $\neq$ algebraically-independent (both use same FK toolbox). The cross-track is structural cross-link, not multiplicative evidence.

Cross-track ratio prediction (Elie observation): $m_e^2 \cdot G / (\hbar c)$ substrate-natural dimensionless $\approx (3\pi/64)^2 \times (G\text{-B substrate factor})^2$ where m_{anchor} cancels algebraically per Planck-mass dimensional analysis.

For Cal #192 cold-read: when Elie's Step 6 + Step 7 land + Lyra's L4 v0.2 m_e numerical lands, verify the cross-track structural identity holds. If numerical m_e closes at Tier 2 STRUCTURAL precision AND numerical G closes at Tier 2 STRUCTURAL precision via the SAME $m_{\text{anchor}} + \ell_B + \hbar_{\text{BST}}$ identification, that's substantive consistency check (not double confirmation).

Joint closure roadmap (Elie estimate): - Week 1: K3 $\hbar_{\text{BST}}$ (Keeper) + L4 mechanism numerical (Lyra) + Step 6.2-6.4 (Elie) - Week 2: numerical closure both lanes + Cal cold-reads + K-audit - ~1-2 weeks parallel multi-CI

If both Lane D and Lane G-B close at Tier 2 STRUCTURAL precision via shared dimensional bridge: substantive cross-validation that substrate-mechanism content is real, NOT pattern-match. Cal #189 + Cal #192 cold-reads combined.

— Keeper. K206 audit framework filed Monday morning. Seven gates explicit. Cal pre-stage observations absorbed. §14 cross-track double-leverage observation added Monday afternoon per Elie Toy 3695 + Keeper observation. Standing for Elie Step 6 + Lyra L4 v0.2 numerical + Keeper K3 lane work + Cal #192 cold-read.